

CENTAURI64

THE MAGAZINE FOR CENTAURI64 USERS

ISSUE 1

WELCOME TO YOUR CENTAURI64!

GET STARTED WITH BASIC
AND WRITE YOUR FIRST PROGRAMS

PRINT
VARIABLES
INPUT

LOOPS
IF...THEN
GOSUB

SOUND
COLOUR
AND MORE!



The image shows a vintage Centauri64 computer system. A CRT monitor displays a BASIC program. Below the monitor is a keyboard with 'CENTAURI64' printed on the right side. To the left of the keyboard are several software boxes, one of which is labeled 'CENTAURI SOFTWARE'. In the foreground, a cassette tape is visible with the label 'CENTAURI64 DUNGEON OF GHOULE'.

```
CENTAURI64 BASIC V1.0  
READY.  
10 PRINT "HELLO CENTAURI64!"  
20 GOTO 10  
  
RUN  
HELLO CENTAURI64!  
HELLO CENTAURI64!  
HELLO CENTAURI64!  
HELLO CENTAURI64!  
HELLO CENTAURI64!
```

COVER TAPE GAME

DUNGEON OF GHOULE
A TEXT ADVENTURE FOR EVERYONE

THE COMPUTER THAT STARTS WITH YOU.

The Centauri64 arrives ready to program. No software package to load. No mysterious setup. Switch on, type an instruction and the machine answers.

PRINT "HELLO"

That is the beginning. What comes next is up to you.

STARTER PACK

£199

64K CENTAURI64

WELCOME TAPE

ISSUE 1 MAGAZINE

1 GAME

BUILT IN

Centauri BASIC • 32 colours • 64K memory • High Resolution and Arcade display modes.

THIS MONTH

Learn BASIC from the blinking cursor to a complete adventure - then alter the cover game yourself.

CENTAURI COMPUTER SYSTEMS

THE HOME COMPUTER FOR PEOPLE WHO WANT TO MAKE THINGS.

CENTAURI64

Welcome to the first issue. This month we begin at the BASIC prompt and finish inside Castle Ghoule. Every listing here has been written for the same Centauri BASIC that you use - no secret magazine version.

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GOTO, NEW, SAVE and LOAD.

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IF...THEN and the first tiny adventure.

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A complete game using everything learned so far.

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Play it, LIST it, understand it, hack it.

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Send us your software. Payment: £2.

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High Resolution drawing and Lunar Rescue.

MEET YOUR CENTAURI64

The blinking cursor is the machine asking what you want to do.

Switch on and BASIC is ready. Simple programs run on the normal screen, directly beneath the editor. You only leave it when a program deliberately changes display mode.

At the prompt, type `PRINT "HELLO"`. With no line number the instruction happens immediately. `HELLO` appears, but the instruction is not stored.

Now add a line number: `10 PRINT "HELLO"`. This time nothing is printed. The Centauri64 has remembered the line as part of a program.

Type `LIST` to see the stored program. Type `RUN` to execute it. BASIC begins at the lowest numbered line and works down.

Programmers commonly number lines 10, 20, 30 rather than 1, 2, 3. The gaps leave room to insert extra instructions later.

IMMEDIATE	STORED	RUN IT
<code>PRINT "HELLO"</code> <code>HELLO</code>	<code>10 PRINT "HELLO"</code> <code>LIST</code> <code>10 PRINT "HELLO"</code>	<code>RUN</code> <code>HELLO</code>

TRY IT `PRINT 2+2` works immediately too. Experiment at the prompt; numbered lines are for the program you are building.

SURVIVAL GUIDE If a running program will not stop by itself, use the normal Centauri64 break control to return to BASIC.

YOUR FIRST PROGRAM

Two lines make a loop. Four commands make programming practical.

```
10 PRINT "MY CENTAURI64 RULES!"  
20 GOTO 10
```

RUN it. Line 10 prints the message. Line 20 sends execution back to line 10. The message repeats until you stop the program.

CHANGE IT

```
10 PRINT "GHOULE IS COMING!"  
15 PRINT "RUN AWAY!"  
20 GOTO 10
```

Typing an existing line number replaces it. A new line number is inserted in the right place.

NEW

clears the current program. Save first if you want to keep it.

LIST

shows the numbered program currently in memory.

SAVE

stores your current program on a Centauri64 tape.

LOAD

brings a saved tape back so you can RUN or LIST it.

GOOD HABIT

Save a copy before making wild changes. Wild changes are encouraged.

GIVE IT A MEMORY

Variables are simply names attached to values.

STORE A NUMBER

```
10 GOLD=10
20 PRINT GOLD
```

CHANGE IT

```
10 GOLD=10
20 GOLD=GOLD+5
30 PRINT GOLD
```

WHOLE NUMBERS

```
10 A=10/3
20 PRINT A

3
```

Think of GOLD as a labelled place in memory. GOLD=10 puts the value 10 there. GOLD=GOLD+5 takes the old value, adds five, then stores the answer.

Centauri BASIC works with whole numbers. Ten divided by three gives 3, not 3.333. For scores, lives, counters and game logic that is often exactly what you want.

Multiplication and division happen before addition and subtraction. Brackets let you change the order: 2+3*4 gives 14, while (2+3)*4 gives 20.

GAME THINKING

A number can remember an event as well as a quantity. Dungeon of Ghoule will use 0 for 'no key' and 1 for 'key found'.

TALK TO YOUR COMPUTER

INPUT turns a program from a speech into a conversation.

```
10 PRINT "WHAT IS YOUR NAME?"
20 INPUT "NAME";N$
30 PRINT "HELLO "+N$
```

A normal variable holds a number. A variable ending in \$ holds text. N\$ remembers whatever the player types. The + on line 30 joins two pieces of text.

BEGIN AN ADVENTURE

```
10 CLS
20 PRINT "WELCOME TO CASTLE GHOULE!"
30 INPUT "WHAT IS YOUR NAME";N$
40 PRINT "WELCOME, "+N$
50 PRINT "THE DOOR CREAKS OPEN..."
```

The player's own answer has become part of the story. Next we need to let an answer change what happens.

TRY IT

Change N\$ to HERO\$. Ask another question and store its answer in a second \$ variable.

REMEMBER

CLS clears the normal BASIC screen. Text programs do not need a separate display mode.

NEXT

INPUT gives us a choice. IF...THEN will let the computer decide what that choice means.

MAKING DECISIONS

IF...THEN is the fork in the road.

```
10 INPUT "HOW MUCH GOLD";GOLD
20 IF GOLD>10 THEN GOTO 100
30 PRINT "YOU CANNOT AFFORD IT!"
40 END
100 PRINT "SOLD!"
```

IF checks a condition. If it is true, THEN tells BASIC what to do. Here, having more than 10 gold sends execution to line 100.

USEFUL COMPARISONS

=	EQUAL
<	LESS THAN
>	GREATER THAN
<=	LESS OR EQUAL
>=	GREATER OR EQUAL
<>	NOT EQUAL

A TWO-CHOICE GAME

```
10 CLS
20 PRINT "YOU STAND BEFORE A DARK DOOR."
40 PRINT "1 OPEN THE DOOR"
50 PRINT "2 RUN AWAY"
60 INPUT "CHOICE";A
70 IF A=1 THEN GOTO 100
80 IF A=2 THEN GOTO 200
90 GOTO 40
100 PRINT "THE DOOR CREAKS OPEN..."
110 END
200 PRINT "YOU RUN HOME!"
210 END
```

The player chooses. INPUT remembers the choice. IF decides where the program goes. That is already enough to make a game.

Dungeon of Ghoule uses the same trick. Its castle is bigger, but the machinery underneath is no more mysterious.

BUILD A TINY ADVENTURE

Use groups of line numbers like rooms on a map.

```
10 CLS
20 PRINT "YOU ARE IN A DARK HALL."
40 PRINT "1 GO TO THE KITCHEN"
50 PRINT "2 GO TO THE CELLAR"
60 INPUT "CHOICE";A
70 IF A=1 THEN GOTO 200
80 IF A=2 THEN GOTO 300
90 GOTO 40

200 CLS
210 PRINT "THE KITCHEN"
230 PRINT "A CAKE SITS ON THE TABLE."
250 PRINT "1 RETURN TO THE HALL"
260 INPUT "CHOICE";A
270 IF A=1 THEN GOTO 10

300 CLS
310 PRINT "THE CELLAR"
330 PRINT "IT IS COLD AND DARK."
350 PRINT "1 RETURN TO THE HALL"
360 INPUT "CHOICE";A
370 IF A=1 THEN GOTO 10
```

BASIC normally runs down through the numbered lines. GOTO changes the route. Keeping locations in separate ranges makes a longer listing easier to understand.

ADD A ROOM!

Add option 3 to the Hall and send it to line 400. Dungeon? Treasure room? Dragon's lair? You decide.

WHY THE GAPS?

Using 200, 300, 400 leaves space for extra lines inside each room later.

LEAVE IT TO CHANCE

RND lets the Centauri64 surprise the player.

WHAT DOES RND DO?

```
10 PRINT RND(2)
20 GOTO 10
```

0 OR 1

RND(2) produces 0 or 1. RND(10) produces a whole number from 0 to 9.

AN EVENT

```
20 R=RND(2)
30 IF R=0 THEN GOTO 100
40 GOTO 200
```

```
100 PRINT "YOU FOUND A GOLD COIN!"
110 END
```

```
200 PRINT "A BAT FLIES PAST!"
210 END
```

Run the program several times. The same listing can produce different events because the computer makes one of the choices.

FROM THE CRYPT

```
696 R=RND(2)
697 PRINT ""
698 IF R=0 THEN GOTO 1200
699 GOTO 1210
```

Those four lines come straight from Dungeon of Ghoulé.

COLOUR YOUR CENTAURI

INK chooses the writing. PAPER chooses the page.

```
10 PAPER 0
20 INK 15
30 CLS
40 PRINT "WELCOME TO MY PROGRAM!"
```

Set PAPER before CLS if you want the display cleared to the new background colour. INK changes the colour used by text that follows.

32 COLOURS

							
0	1	2	3	4	5	6	7
							
8	9	10	11	12	13	14	15
							
16	17	18	19	20	21	22	23
							
24	25	26	27	28	29	30	31

HELP! WHAT HAVE I DONE?

Experimenting with colours is half the fun. If the BASIC screen ends up in settings you no longer want, type RESET at the prompt.

NEW IS DIFFERENT

NEW clears the program in memory. RESET restores machine settings. They solve two different problems.

RESET

GUESS MY NUMBER

Your first complete type-in game - using nothing you haven't already learned.

Type the listing carefully, then RUN it. The Centauri64 chooses a secret number from 0 to 9 and gives you higher/lower clues until you find it.

```
10 PAPER 0
20 INK 15
30 CLS
40 SECRET=RND(10)
50 PRINT "GUESS MY NUMBER!"
60 PRINT ""
70 PRINT "I AM THINKING OF A NUMBER"
80 PRINT "FROM 0 TO 9."
90 PRINT ""
100 INPUT "YOUR GUESS";A
110 IF A=SECRET THEN GOTO 200
120 IF A<SECRET THEN GOTO 160
130 PRINT "TOO HIGH!"
140 GOTO 100
160 PRINT "TOO LOW!"
170 GOTO 100
200 PRINT ""
210 INK 7
220 PRINT "YOU GOT IT!"
230 END
```

WHAT'S HAPPENING?

Line 40 chooses the secret. Line 100 asks for a guess. Lines 110 and 120 compare it with the secret. The GOTOs keep the player guessing until the answer is correct.

TYPE IT - DON'T PASTE IT

The point of a type-in is that your fingers and brain meet every line. If you make an error, LIST the program and hunt it down.

NOW MAKE IT YOURS

The best type-in is the one you stop copying.

MAKE IT HARDER

```
40 SECRET=RND(100)
80 PRINT "FROM 0 TO 99. "
```

Two changes make the game much tougher.

MAKE IT MEANER

Remove the TOO HIGH and TOO LOW clues. Now the player only learns whether the guess is correct.

ADD A SCORE

Can you count how many guesses the player makes? You already know enough BASIC to try.

A type-in is not a museum piece. Change the title. Change the range. Change the colours. Add messages. Break it, LIST it, repair it. When you are happy, SAVE your version to your own tape.

CHALLENGE

Can you make the computer choose from 1 to 10 instead of 0 to 9? Think about what RND(10) actually returns.

SAVE IT

Give your version a new tape name. You have made a real Centauri64 game.

DUNGEON OF GHOULE

DARE YOU ENTER CASTLE GHOULE?

High above the village stands Castle Ghoule. Nobody has entered its gates in years - or at least, nobody who has returned. Tonight the gates are open.

Explore the ruined castle, search its forgotten rooms and discover the secret guarded by the ghost in the tower. Choose carefully. Not everything buried beneath Castle Ghoule wants to be found.

HOW TO PLAY: type the number beside your choice and press RETURN. Your decisions decide where the story goes.

The most important thing about this month's cover game is not the treasure. It is that you can LOAD it, LIST it and see every instruction that makes the adventure work.

ON THIS MONTH'S COVER TAPE

DUNGEON OF GHOULE

LOAD IT • RUN IT • LIST IT • CHANGE IT

BEFORE YOU HACK SAVE your own copy first. Then the castle is yours.

HOW DOES IT WORK?

There is no secret adventure command. You already know the machinery.

```
150 PRINT "1 ENTER THE CASTLE"  
160 PRINT "2 RUN AWAY"  
170 INPUT "CHOICE";A  
180 IF A=1 THEN GOTO 200  
190 IF A=2 THEN GOTO 900
```

Dungeon of Ghoule prints choices, waits for a number, then uses IF and GOTO to decide where execution continues.

THE CASTLE IN LINE NUMBERS

200 GREAT HALL

400 TOWER

500 CELLAR

600 CRYPT

700 GHOST

1000 TREASURE ROOM

Keeping locations in separate groups of line numbers makes a long listing easier to understand. It also leaves room to insert new instructions.

Notice how the Great Hall acts as a hub. Several choices send the player out to other locations; those locations can send the player back.

Once you see the listing as a collection of small sections rather than one enormous program, the adventure becomes much less intimidating.

THE SILVER KEY

One variable gives the adventure a memory.

AT THE START

```
60 KEY=0
```

IN THE CELLAR

```
570 KEY=1  
575 PRINT "YOU FOUND A SILVER KEY!"
```

AT THE GHOST

```
730 IF KEY=1 THEN GOTO 800
```

That's the whole trick. KEY=0 means the player has not found the key. KEY=1 means they have. When the player meets the ghost, the program checks the value.

A variable does not need to represent a visible number. It can remember whether something happened earlier. That gives the game a history.

The same idea can represent a door switch, a rescued prisoner, a treasure already collected, a boss defeated or a conversation already heard.

THE LESSON

You do not need a special inventory command to begin making adventures. A handful of well-chosen variables can remember a surprising amount about the player's journey.

EXPECT THE UNEXPECTED

The Crypt does not behave the same way every time.

```
696 R=RND(2)
697 PRINT ""
698 IF R=0 THEN GOTO 1200
699 GOTO 1210
```

One result sends bats bursting from the walls. The other gives the player a whispered clue about the silver key.

DUNGEON OF GHOULE IS REALLY JUST...

PRINT	describe the world
INPUT	ask what the player wants
IF	make decisions
GOTO	move around the adventure
VARIABLES	remember what happened
RND	add surprise
INK / PAPER / CLS	set the mood

NO SECRET ENGINE The cover game is made from the same BASIC you've been learning. LOAD it, LIST it, read it and change it.

HACK THE DUNGEON!

Don't just play the cover tape. Make it yours.

1 • EASY

REWRITE THE GHOST

Change the ghost's dialogue. Make it frightening, friendly or ridiculous.

2 • EASY

CHANGE THE TREASURE

Gold is predictable. What else could be hidden in the secret room?

3 • MEDIUM

CHANGE THE CRYPT

Replace the bats or whispered clue with your own random event.

4 • MEDIUM

ADD ANOTHER CHOICE

Give one existing room a third option and send it somewhere new.

5 • HARD

BUILD A NEW ROOM

Create a new location around line 1300 and connect it to Castle Ghoulé.

SAVE YOUR VERSION

Give your altered game a new name and save it to your own tape. Keep the original cover tape untouched so you can always start again.

READER CHALLENGE

YOUR FIRST PROGRAM

WE WANT YOUR SOFTWARE!

Have you written something for your Centauri64? Send it in. It does not need graphics. It does not need sound. It does not even need to be a game.

We are looking for something that proves you have started programming your new computer. Save it to a tape, give it a description and author, then submit it through the Reader Challenge.

Even a tiny looping HELLO program can qualify. This is your first step, not your final exam.

TO ENTER

1. Write a Centauri BASIC program.
2. Save it to a tape.
3. Add description and author.
4. Submit it to the challenge.

PAYMENT

£2

```
10 PRINT "HELLO"  
20 GOTO 10
```

YES, SIMPLE COUNTS The point is to get started, finish something and send it. There will be harder challenges later.

GRAPHICS!

YOU'VE MADE IT TALK. NOW MAKE IT DRAW.

Issue 2 opens the Centauri64's High Resolution display. Plot points. Draw lines. Build rectangles and circles. Add colour. Then combine those ideas into your first graphical game.

YOU'LL LEARN

MODE 1 • PLOT • LINE • RECT • CIRCLE • movement •
timing • simple graphical game loops

ON THE COVER TAPE

LUNAR RESCUE

YOUR FIRST GRAPHICAL
CENTAURI64 GAME

```
10 MODE 1
20 PLOT 160,120
30 LINE 20,20,300,200
```

ON SALE NEXT MONTH